



Enhanced energy absorption of auxetic cementitious composites with polyurethane foam layers for building protection application

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ABSTRACT

Cementitious composites with impact energy absorption capabilities can have wide applications in protective structures. This paper presents an experimental investigation on the flexural response and energy absorption of auxetic cementitious composites reinforced with polyurethane foam. In total, ninety (90) composite specimens were prepared from polymer cement mortar overlayed with either carbon fibre fabric or auxetic fabric and polyurethane foam insertions in the upper and lower regions. These specimens were tested under flexural loads using a four-point loading setup at varying loading speeds from 1 mm/min to 300 mm/min. Digital image correlation method was employed to study the failure modes, displacement profiles, load-displacement curves, and energy absorption characteristics. The auxetic fabric overlayed composites exhibited increased energy absorption and ductility without any signs of debonding compared to the commonly used carbon fibre overlayed composite specimens. Polyurethane foam insertions in the upper part of the composites prevented splitting of the specimen by reducing the crack widening in the compression zone of the cementitious matrix. Additionally, an increase in loading rate improved the energy absorption capacity of the auxetic fabric cementitious composites compared to carbon fibre cementitious composites. Findings from this study will have applications for protecting structures that are vulnerable to impacts.

1. Introduction

Cementitious renders used as exterior coatings for protecting underlying structures that are vulnerable to impact, need to have adequate capacity to resist the impact. Standard mortars tend to be brittle, making it challenging for the render to provide sufficient resistance [1–4]. The use of fibres, such as steel [5,6], carbon [7–10], and basalt [11], can improve the strength of the render. Fibres such as wool [12,13], cotton [14,15], wood, hems [16], coconut [14,15] and others [15] are commonly used to reinforce the cementitious renders to improve their structural and fire performance and for protection of underlying structures [17–19]. However, in applications where the render is impacted by flying debris during cyclonic events, blasts, vehicular crashes or similar disasters, its energy absorption property becomes a key parameter in protecting the underlying structure [3,20–22]. To improve the energy absorption capacity of rendering composites, both strength and flexibility/plastic deformations with ductile behaviour must be achieved [23,24].

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Previous researchers [10,25,26] analysed the performance of carbon fibre (CF) fabric in structural rehabilitation applications. These CF fabrics exhibited very high tensile strength (in the order of ~ 3000 MPa to ~ 4000 MPa), low density and high stiffness but were found to be vulnerable to delamination or debonding during impact and blast loading [3,27–30]. The delamination and debonding phenomenon of the CF fabric significantly reduces the flexibility of the composites, which ultimately minimizes the energy absorption capability of the CF-strengthened composite structure. To overcome this limitation, researchers [10,31,32] have developed auxetic cementitious composites (ACC) with a negative Poisson's ratio (NPR), which enabled high energy absorption. NPR behaviour means expansion or contraction respectively in the transverse direction when an auxetic material is stretched or compressed along its longitudinal axis. This behaviour also enhances the bonding characteristics of auxetic materials. The use of auxetic fabric in ACC has been shown to improve the debonding and energy absorption characteristics of the composites compared to traditional CF-strengthened composites [10,32–34]. The effectiveness of auxetic fabric (AF) has also been studied for different applications that require high energy absorption, such as crashworthiness of road vehicles [35–37] and fall protection in biomedical engineering [38]. However, their applications in civil engineering [3,30,39–43] are relatively new and more research is needed to fully understand the potential of ACC in protecting buildings and structures from impact and blast loadings.

In previous studies [10,31,44], the AF fabric with an NPR of -2.1 was used to prepare ACC samples to test under bending. The AF fabric had a significantly lower tensile strength of 50 MPa compared to the CF fabric, which was around 3000 MPa. However, flexural tests of these ACC specimens showed that the composites with AF fabric had significantly enhanced bonding and energy absorption characteristics compared to the traditional CF fabric-strengthened composites. The ACC specimens showed 2.6 times higher flexural strength and 25% higher energy absorption compared to the CF fabric-strengthened specimens under bending when tested at a rate of 150 mm/min [3,10]. This study advances the knowledge on flexural behaviour of ACC by employing an advanced AF fabric weft with basalt fibres of high tensile strength (about ~ 400 MPa [45]) in the transverse direction as an overlay for cementitious matrix. This is a second-generation auxetic fabric (AF 2.0) with enhanced mechanical properties such as durability, high energy absorption capacity, and fire resistance.

Polyurethane foam layers have been inserted to enhance the energy absorption capacity of the ACC specimens constructed using AF 2.0 fabric. According to the literature [46–49], recent advances have been made in developing lightweight flexible sandwiched composites composed of flexible polyurethane (PU) foam filled in the substrate core with one or two laminates as face sheets. These composites can potentially be used in aeroplanes, impact-resistant construction, and other applications [40,50] due to the various advantages of PU foam, such as high production efficiency, high energy absorption, and ease of manufacturing. The energy absorption of these lightweight flexible sandwiched composites can be enhanced by overlaying laminates in the tension zone of the composite. This additional overlay will improve the bending modulus and strength, while the filled PU foam enhances flexibility and reduces the weight of the composite [46,49,51]. Similar sandwich composites can be developed with AF 2.0 fabric and cementitious matrix for impact resistance applications in civil structures.

This paper presents the manufacturing and characterisation of flexible sandwiched composite specimens containing AF 2.0 fabric overlays and PU foam as inserts in the cementitious matrix at different thickness levels. The overlays and inserts were positioned in the wet cementitious matrix without the use of adhesives. The bond between the cementitious matrix and the overlay fabric, as well as the inserted polyurethane foam layers developed naturally through cement hydration during curing. The specimens were tested under bending at three distinct loading rates (1, 150, and 300 mm/min), and the responses of the composites were measured using the digital image correlation (DIC) method. A total of ninety (90) specimens were manufactured and tested. The different positioning of the PU foam inserts in the composite and the AF 2.0 fabric overlay were trialled to study the energy absorption performance of the composites. Additionally, the effectiveness of the auxetic cementitious composites in terms of energy absorption was also compared with different textiles-reinforced cementitious composites available in the literature and reported in section 3.4. A comparative study was conducted to propose a highly ductile cost efficient auxetic textile reinforced cementitious composite render for protecting masonry buildings vulnerable to impacts or vehicular crashes.

2. Experimental programme

Composite specimens of dimensions 240 mm long $\times$ 45 mm wide $\times$ 20 mm deep were fabricated and tested. A layer of AF 2.0 fabric was overlayed at the bottom of the specimens, whereas a single layer of PU foam was inserted at two different depths, i.e., 5 mm and 15 mm from the top surface of the specimens. Benchmark specimens were also prepared in four different configurations (i) plain mortar samples without overlays and inserts, (ii) mortar samples with auxetic fabric overlays only, (iii) mortar samples with PU foam inserts only, and (iv) plain mortar samples with overlays and inserts. After 14 days of air curing, the specimens were tested under four-point bending at three different rates of deformation-controlled loading (1, 150 and 300 mm/min). It must be mentioned that with the insertion of AF 2.0 fabric (1 mm thick) or CF fabric (0.5 mm thick) or PU foam (5 mm thick) in the composite specimens of overall thickness of 20 mm, the volume content of the cementitious matrix was reduced accordingly.

Digital Image Correlation (DIC) method was used to measure cracking, crack widening and debonding in specimens. Failure modes, load-displacement curves, ultimate load, and energy absorption for all specimens were measured. The 2-D surface used for recording and analysing during the DIC setup was parallel to the thickness of the specimen, as shown in Fig. 1. The specimens' overlayed fabric, foam insert location, and orientation within the 2-D surface are also presented in Fig. 1(b).

2.1. Material properties

This section reports the properties of the four constituent materials: (a) cementitious matrix, (b) Carbon Fibre Reinforced Polymer (CF) fabric, (c) second-generation auxetic fabric (AF 2.0), and (d) Polyurethane foam (PU foams).

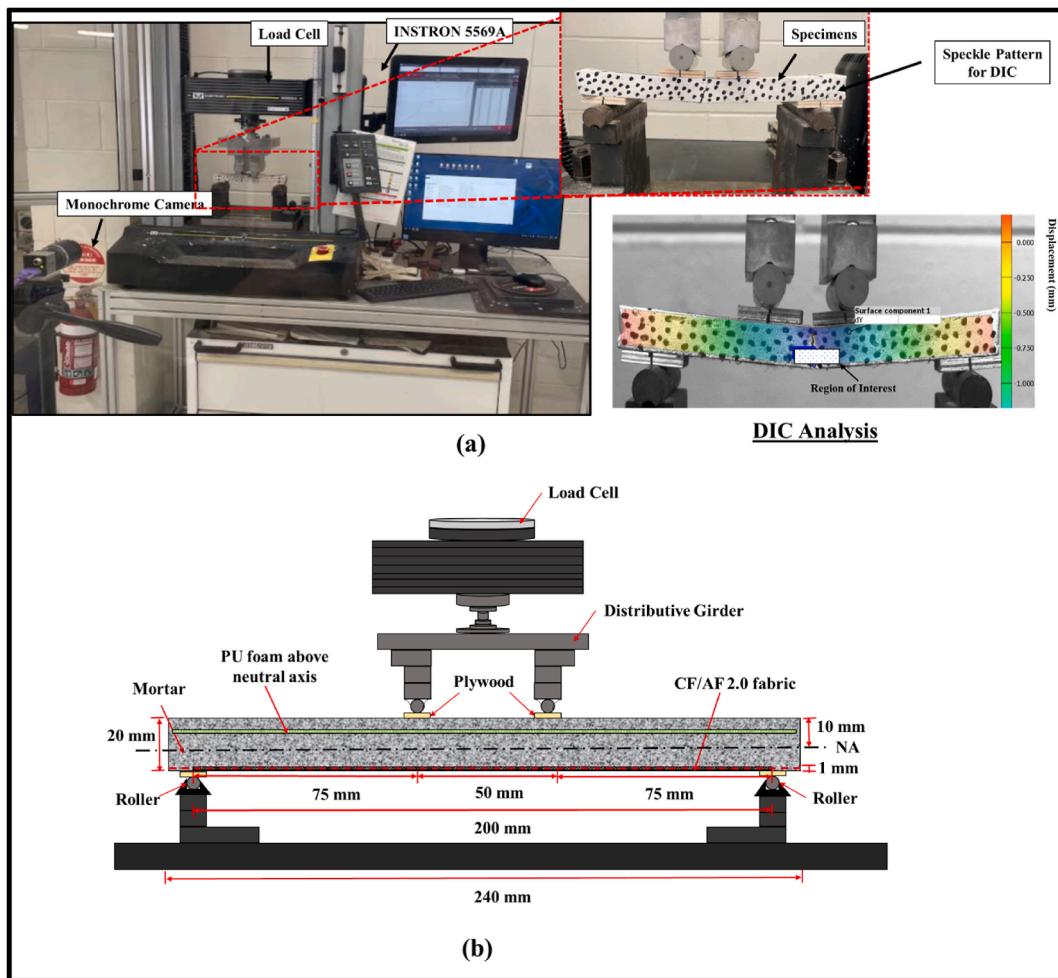


Fig. 1. (a) Testing setup for the composite specimens and (b) 2-D surface selected for the DIC measurement during flexural loading.

2.1.1. Cementitious matrix: Plain Acrylic mortar (PM)

Cementitious matrix was prepared using a commercially available acrylic mortar render satisfying the requirements of class R4 of the European standard EN 1504-3. The pre-mixed acrylic mortar was blended with water in the proportion of 1:0.18 by weight and was moulded into cylindrical specimens of 50 mm diameter and 100 mm height to test under compression. The mean compressive strength of the mortar mix was measured as 5 MPa by testing three identical specimens. This acrylic mix render was previously used as a thin layer coating for masonry structural walls [15,52]. The advantage of using acrylic rendering was that it could also be applied to various substrates, including painted walls, concrete, timber, fibre cement and so on.

2.1.2. Carbon fibre reinforced polymer (CF) fabric

The Carbon Fibre (CF) fabric of 0.5 mm thick was used as a comparison material for the AF 2.0 fabric in this study. The CF fabric was available with the fibres aligned unidirectionally, which provided a meaningful comparison with the AF 2.0 fabric, as shown in Fig. 2(a). The average tensile strength and elastic modulus of the single-strand flax yarns extracted from the fabric were 3200 ± 300 MPa and 225 ± 50 GPa, respectively. The nominal cross-section area and the strain to failure of the CF fabric provided by the supplier were $129 \text{ mm}^2/\text{m}$ and 1.56% respectively. These values show that the CF fabric has high tensile strength and stiffness compared to the AF 2.0 fabric, but it is more prone to delamination and debonding under impact and blast loadings.

2.1.3. Second-generation auxetic (AF 2.0) fabric

The auxetic fabric AF 2.0 used in this research was an upgraded version of the unidirectional woven Auxetic fabric (AF) used by the authors in their previous research [10,31]. The second-generation woven auxetic fabric is made from helical auxetic yarn of circular cross-section in the axial direction and basalt fibres of diameter $10\text{--}20 \text{ }\mu\text{m}$ woven in a transverse direction as shown in Fig. 2(b). The nominal cross-section area of the auxetic fabric was $250 \text{ mm}^2/\text{m}$ per unit width. The fabric was designed and woven by a commercial supplier in the United States. It was tested under tension in an INSTRON 5566 machine at QUT, Australia. The size and shape of the 1 mm thick auxetic fabric cut for testing are shown in Fig. 2(b).

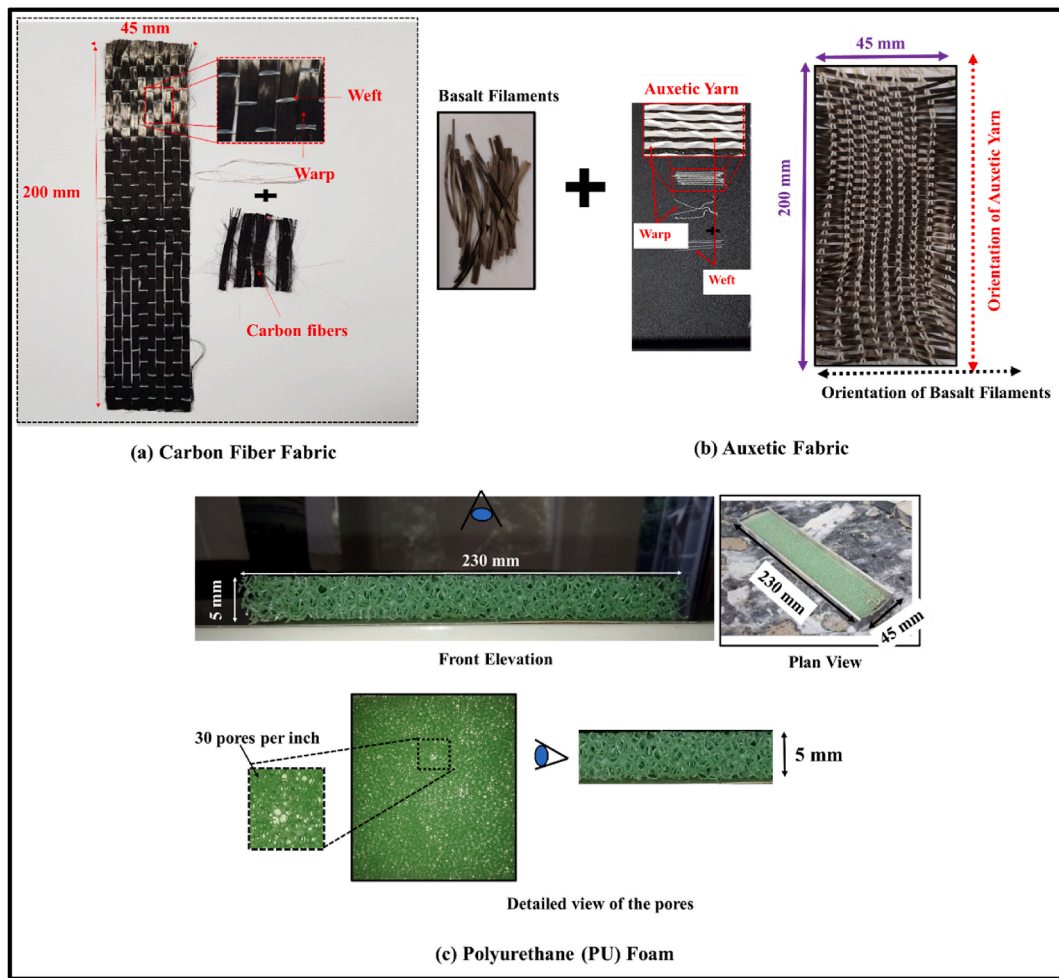


Fig. 2. Details of (a) Carbon Fibre Reinforced Polymer (CF) Fabric (b) Second-generation Auxetic (AF 2.0) fabric and (c) Polyurethane (PU) Foam with open pores detailing.

Three specimens were tested under a uniform displacement rate of 1 mm/min and the results were analysed using the DIC method. The specimens exhibited different strengths based on the axial loading direction and the orientation of the auxetic yarn and basalt filaments. Tensile strength of ~ 100 MPa and NPR of -2.1 were obtained for AF 2.0 for loading parallel to the auxetic fibres, whereas when loaded in a transverse direction (parallel to the basalt filaments), tensile strength of ~ 200 MPa was measured.

2.1.4. Polyurethane (PU) foam

The Polyurethane foam (PU foam) of 5 mm thickness used in this research was a conventional inexpensive open-cell 30 pores per inch (ppi) PU foam (shown in Fig. 2(c)) purchased from Western Australia. The density of the 30 ppi foam was 28.85 ± 1.8 kg/m³. The compressive strength, Poisson's ratio, and complete stress-strain behaviour of the PU foam were determined using a 1 kN INSTRON 5566 machine using the Digital Image Correlation (DIC) method [4]. The stress-strain curve was reported by one of the authors of this paper in Ref. [4]. From the curve, the average positive Poisson's ratio was calculated as 0.2, and the average tensile strength was 0.06 MPa.

2.2. Specimen preparation and strengthening technique

In total ninety (90) specimens of mortar composites were prepared with CF/AF 2.0 overlays and PU insertion at different thickness locations. The prepared specimens were tested under 4-point bending at different displacement rates (1, 150 and 300 mm/min). Rectangular C-shaped channel (240 mm long $\times$ 40 mm wide $\times$ 20 mm deep) moulds were used as formwork. For PU foam insertion, the moulds were marked at appropriate locations on the sides which guided the placements of the PU foam during the filling of mortar in the mould. For casting the specimens, aluminium channel sections were used to prepare the mould of specified dimensions as shown in Fig. 3(a). The inside surfaces of the moulds were greased, and the AF 2.0 fabric or CF fabric was placed at the bottom as shown in Fig. 3(b) and (c). A 5 mm thick PU foam layer was inserted at the desired location above or below the neutral axis. The mould was then filled with more polymer cement mortar to the top as shown in Fig. 3(d). The prepared specimens are presented in Fig. 3(e).

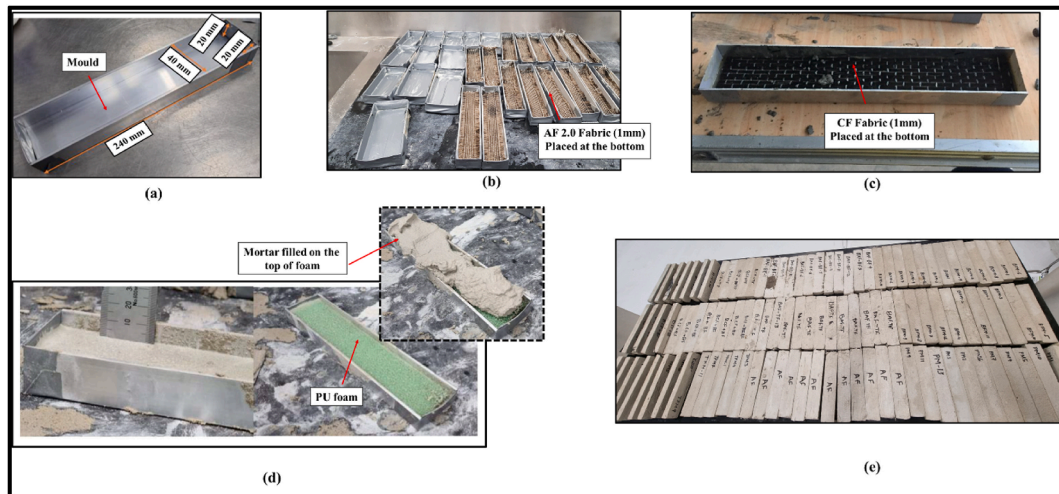


Fig. 3. Specimens preparation (a) preparation of mould, (b) Auxetic fabric placed in the mould, (c) CFRP fabric, (d) PU foam insertion and (e) prepared specimens.

Table 1 lists the identifier used for each specimen; the general identifier is AA-BB-CCC-D. The first two letters, “AA” stand for the type of specimen-plain mortar (PM)/auxetic fabric (AF)/CFRP fabric (CF). The second two letters, “BB” stand for the location of inserted foam (F) – none NO/foam inserted above the neutral axis at 5 mm from the top surface (TF)/foam inserted below the neutral axis at 15 mm from the top surface (BF). The subsequent three letters “CCC” stand for the rate of loading in mm/min- “001”/“150”/“300”. Please note that a maximum of five specimens were fabricated in most groups and some groups had only three specimens.

Table 1
Nomenclature and description of the developed specimens.

Specimen Name	Explanation	Number of Specimens
Four-Point Bending Test		
<i>Group-I (AF 2.0 and PU composites at a displacement rate of 1 mm/min)</i>		
PM-NO-001	Plain Mortar	3
PM-TF-001	Plain Mortar + PU insert above the neutral axis	3
PM-BF-001	Plain Mortar + PU insert below the neutral axis	3
CF-NO-001	Plain Mortar + CF overlay	3
CF-TF-001	Plain Mortar + CF overlay + PU insert above the neutral axis	3
CF-BF-001	Plain Mortar + CF overlay + PU insert below the neutral axis	3
AF-NO-001	Plain Mortar + AF 2.0 overlay	3
AF-TF-001	Plain Mortar + AF 2.0 overlay + PU insert above the neutral axis	3
AF-BF-001	Plain Mortar + AF 2.0 overlay + PU insert below the neutral axis	3
<i>Group-II (AF 2.0 and PU composites at a displacement rate of 150 mm/min)</i>		
PM-NO-150	Plain Mortar	3
PM-TF-150	Plain Mortar + PU insert above the neutral axis	3
PM-BF-150	Plain Mortar + PU insert below the neutral axis	3
CF-NO-150	Plain Mortar + CF overlay	3
CF-TF-150	Plain Mortar + CF overlay + PU insert above the neutral axis	4
CF-BF-150	Plain Mortar + CF overlay + PU insert below the neutral axis	4
AF-NO-150	Plain Mortar + AF 2.0 overlay	3
AF-TF-150	Plain Mortar + AF 2.0 overlay + PU insert above the neutral axis	4
AF-BF-150	Plain Mortar + AF 2.0 overlay + PU insert below the neutral axis	3
<i>Group-III (AF 2.0 and PU composites at a displacement rate of 300 mm/min)</i>		
PM-NO-300	Plain Mortar	3
PM-TF-300	Plain Mortar + PU insert above the neutral axis	3
PM-BF-300	Plain Mortar + PU insert below the neutral axis	3
CF-NO-300	Plain Mortar + CF overlay	4
CF-TF-300	Plain Mortar + CF overlay + PU insert above the neutral axis	3
CF-BF-300	Plain Mortar + CF overlay + PU insert below the neutral axis	4
AF-NO-300	Plain Mortar + AF 2.0 overlay	4
AF-TF-300	Plain Mortar + AF 2.0 overlay + PU insert above the neutral axis	3
AF-BF-300	Plain Mortar + AF 2.0 overlay + PU insert below the neutral axis	3

2.3. Instrumentation and test setup

The prepared cementitious composite specimens were tested using a 50 kN INSTRON 5569 machine with 0.0001 precision under four-point bending tests (as shown in Fig. 1(a)). All the specimens were tested at 1 mm/min, 150 mm/min and 300 mm/min to investigate the effect of loading rate.

Digital Image Correlation (DIC) system was used to measure the specimen deformations under the four-point bending tests with the region of interest for measurement as shown in Fig. 1(a). The DIC system comprises an 8-bit 4-megapixel monochrome camera, tripod and DIC analysis software. As shown in Fig. 1(a), the monochrome camera was positioned perpendicular to the tested specimen and captured the images at 160 frames per second. The captured images were later analysed to measure the displacements of specimens over time at the point of interest under the four-point loading. The applied flexural loads over time were obtained from the INSTRON machine. The exported flexural loads from the INSTRON machine and measured specimen displacements from the software were plotted against each other to obtain force-displacement graphs. The behaviour of each cementitious composite type was investigated in terms of stiffness and ductility. Detailed results of the force vs displacement of each sample are presented in Section 3.

3. Results and discussion

3.1. Failure behaviour of the composites

Fig. 1(a) shows a typical displacement plot of coloured fringes obtained from the DIC plot. The following subsections present the failure modes of the composite specimens.

3.1.1. Plain Acrylic mortar containing NO inserts (PM-NO)

The specimen in Fig. 4 failed suddenly at the mid-span under the 4-point bending test at a loading rate of 300 mm/min, with a single crack forming in the tension zone of the bottom section. The same failure mode was observed at slower loading rates but with a decrease in peak load. The specimen tested at 1 mm/min had a peak load of 183.36 N, compared to 206.21 N at 300 mm/min, due to the dynamic loading rate effect. The dynamic effect of plain mortar was described earlier by the authors in Refs. [10,53,54].

3.1.2. Plain Acrylic mortar – CF fabric overlay composites (CF-NO)

The composite specimens with CF fabric overlays, as shown in Fig. 5, exhibited tensile cracks emanating from the location of the tensile fabric and propagating towards the compression zone of the specimen at a loading rate of 300 mm/min. The specimen failed due to the widening of the dominant crack. The study shows that the different loading rates of 1 mm/min and 150 mm/min did not affect the failure modes of the composite specimens. Additionally, the specimen also exhibited rupture of the fabric due to kinking failure of the specimen.

3.1.3. Plain Acrylic mortar – AF 2.0 fabric overlay composites (AF-NO)

The flexural failure mode of the composites with AF 2.0 fabric overlays loaded at 300 mm/min is shown in Fig. 6. Two failure modes were commonly observed for the AF 2.0 overlayed composites when tested under 4-point bending as shown in Fig. 6(a). The AF 2.0 fabric overlays enhanced the flexible behaviour of the composite by preventing sudden failure. The failure mode exhibited no debonding at the interface between AF 2.0 fabric and mortar, but flexural wide cracks/openings were observed in the tension face of the overlayed composite specimen. The non-debonding of the fabric was due to the negative Poisson's ratio effect of the fabric which significantly enhanced the performance of composite specimens in terms of capacity and flexibility under flexural loading. This effect was discussed in more detail in an article [10] by the authors.

In summary, AF 2.0 overlayed composite specimens showed a 36% higher peak load when tested at a 300 mm/min loading rate than at 1 mm/min. The cracking observed in the specimen was initiated within the mortar thickness at the tension face and propagated to the top compression face, resulting in a wide opening throughout the mortar thickness. The composite specimen lost its integrity on the compression face, but the bottom face of the mortar remained attached to the fabric, allowing it to take a tensile load due to its non-debonding characteristics.

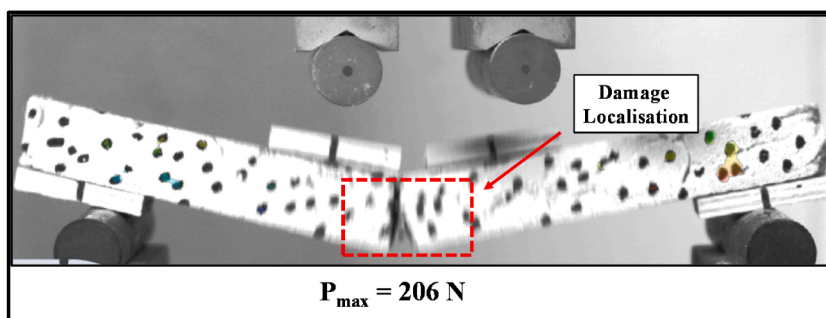


Fig. 4. Failure of Plain Acrylic mortar (PM-NO) (control specimen) at 300 mm/min.

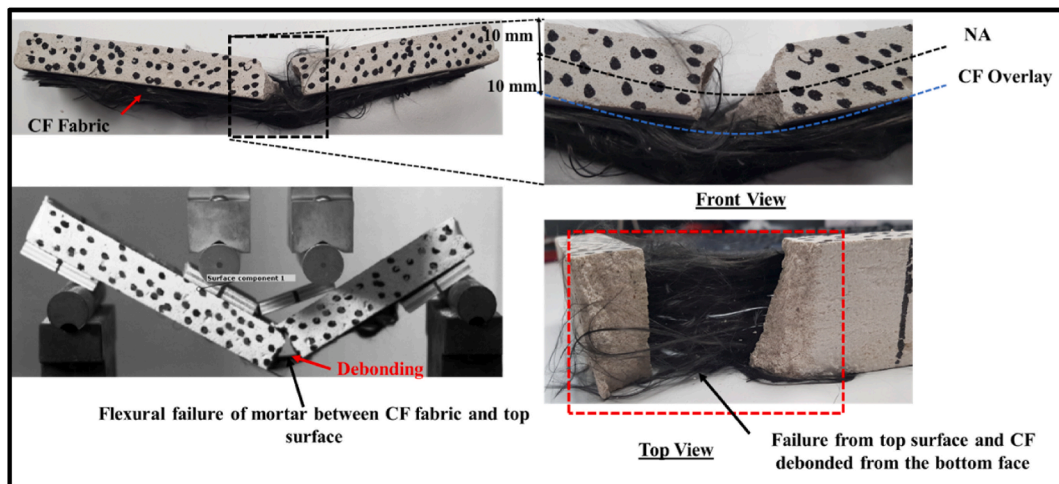


Fig. 5. Failure modes of CF fabric overlayed composite (CF-NO) at 300 mm/min.

3.1.4. Plain Acrylic mortar – AF 2.0 fabric overlay + PU foam insert (AF-TF and AF-BF)

The PU foam was inserted in the specimen as an energy absorption device which significantly enhanced the energy absorption, the load-carrying capacity, and the stability of the damage propagation process. The effect of the foam insertion based on the location across the thickness of the mortar (as shown in Fig. 7(a)) was studied first to determine the best position to enhance composite performance.

The PU foam was inserted at two different locations in the composite specimen: above the neutral axis (about 5 mm deep from the top surface) and secondly, below the neutral axis (about 5 mm above the bottom surface of the specimen) as shown in Fig. 7(a). Results show that the failure modes were significantly influenced with the insertion of PU foam in the AF 2.0 overlayed specimens (AF-TF or AF-BF) when compared to the AF 2.0 overlayed specimen (AF-NO) (presented in Fig. 6(c)). The inserted foam acts as a connector at its location in the specimen, which prevents the splitting of the specimen into two or multiple parts. The failure mode of PU insertion above the neutral axis is shown in Fig. 7(b). Cracks were generated and widened up in the mortar section. However, complete splitting of the specimen from the top compression face did not occur as compared to the AF 2.0 overlay composites failure mode as shown in Fig. 6(c). When the PU foam was inserted below the neutral axis, splitting of the mortar in the top compression zone was not prevented as shown in Fig. 7(c). From the investigation of the failure modes, it is therefore obvious that the presence of the PU foam above the neutral axis was more effective to prevent the complete splitting of the specimen under flexural test.

3.1.5. Plain Acrylic mortar – CF fabric overlay + PU foam insert (CF-TF or CF-BF)

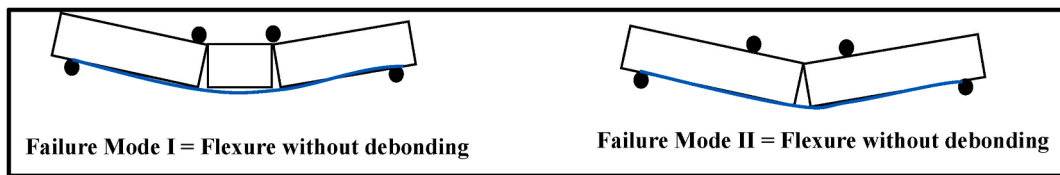
The configurations of the specimens with CF overlays and foam insertions were similar to the composites with AF fabric overlays as shown in Fig. 7(a). The flexural failure modes of the PU foam insert above (top) and below (bottom) the neutral axis of the CF overlayed composite specimens under 300 mm/min loading are shown in Fig. 8(a) and (b) respectively.

Flexural cracks and a wide opening in the mortar (as shown in Fig. 8(a)) within the tension face of the PU insert (above the neutral axis) in the CF overlayed composite specimen were observed as in the AF overlayed composites (Fig. 7(b)). The failure modes exhibited debonding and rupture of the fabric, which are different from the composites overlayed with the AF 2.0 fabric. However, the foam insertion in the top effectively held the mortar even though cracks developed throughout the thickness of the specimen. The bonding between the PU foam and the highly elastic nature of the PU foam prevented the widened crack from propagating from the tension face to the top of the compression face, as shown in Fig. 8(a).

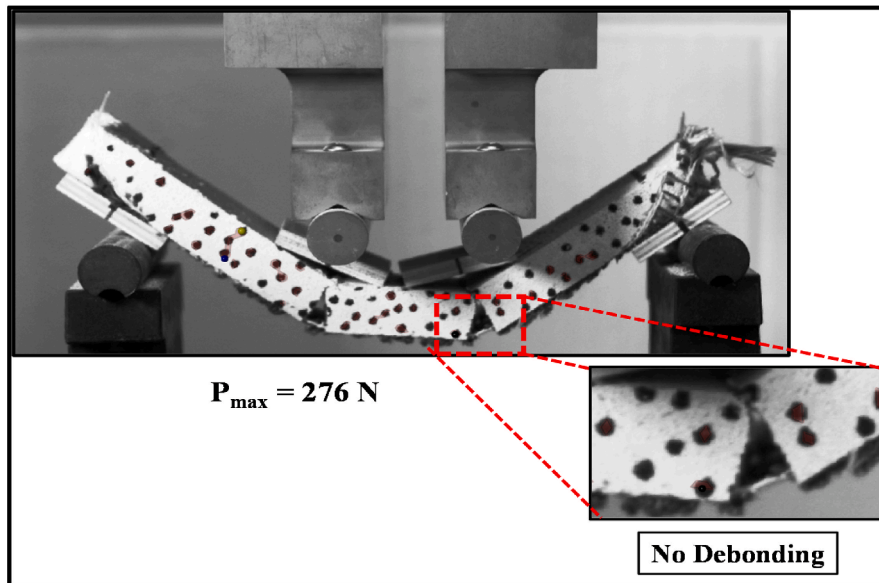
The flexural failure of the PU foam inserts at the bottom of the CF overlayed composites as shown in Fig. 8(b), was slightly different compared with the same configuration used in AF 2.0 overlayed composite specimens (Fig. 7(c)). The inserted PU foam fractured due to the localised stress developed when the debonding and rupture of the CF fabric occurred. As a result, the specimen was utterly fragmented into two pieces, as seen in Fig. 8(b).

Typical failure modes observed for the 90 specimens with different configurations are schematically presented in Fig. 9. Failure modes are grouped for the PM (PM-NO), PM + PU inserts (PM-TF and PM-BF), CF overlayed (CF-NO) and CF + PU inserts (CF-TF and CF-BF), AF 2.0 overlayed (AF-NO) and AF 2.0 + PU inserts (AF-TF and AF-BF).

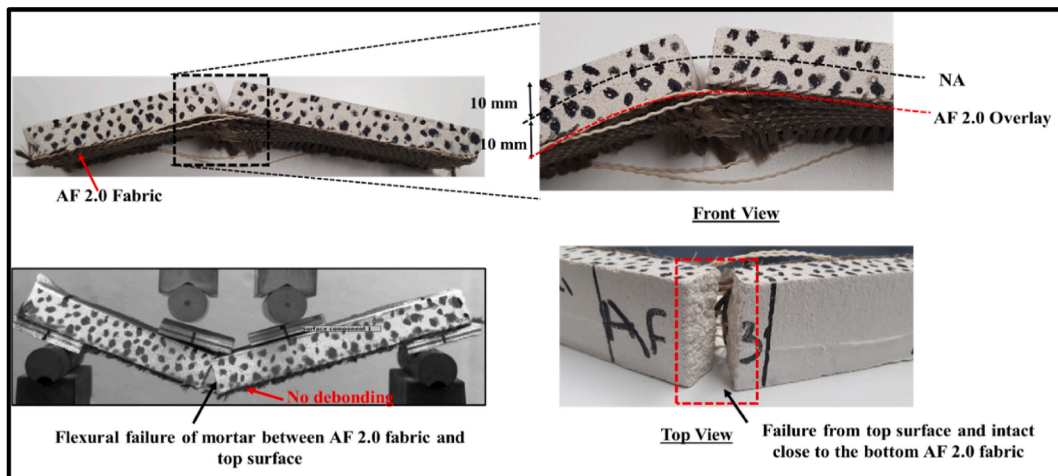
The PM specimens failed due to the formation of a single, wide crack either at mid-span or near one of the loading points as shown in Group I (Fig. 9 (a) and (b)). Insertion of PU foam in the top compression zone of the PM specimen prevented the crack from widening up to the top face and allowed the specimens to remain integral till the end of testing as shown in Group II (Fig. 9(c)). However, the insertion of PU foam below the neutral axis prevented the cracks widening at its location with a major failure occurring due to cracks widening up to the top compression face (Fig. 9(d)). The CF overlayed specimens failed due to the wide cracks opening in association with debonding and fracturing of the fabrics as shown in Group III (Fig. 9(e)). Insertion of PU foam in the top compression zone of the CF overlayed specimen (as shown in Fig. 9(f); Group IV) prevented the crack from widening up to the top face, which allowed the specimens to remain integral till the end of testing. However, insertion of PU foam below the neutral axis prevented the



(a) Modes of Failure in AF 2.0 overlaid composite specimens (AF-NO) during 4-point bending test

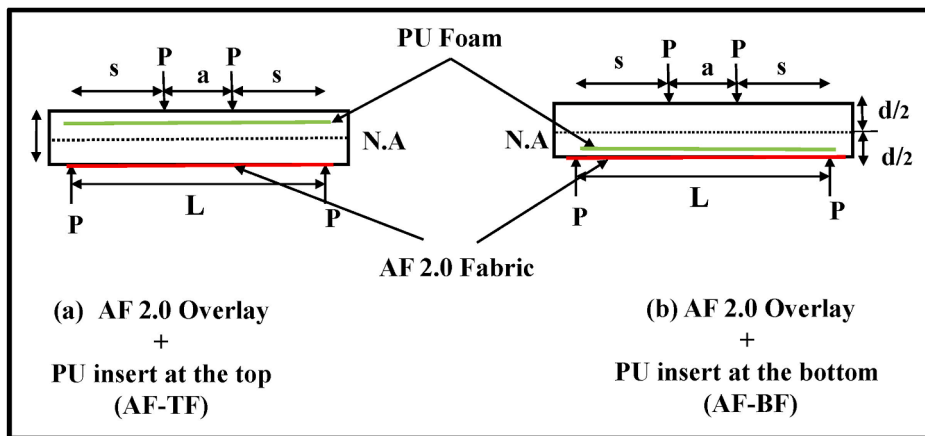


(b) Failure mode I of AF 2.0 overlaid composites (AF-NO) under 4 point bending test

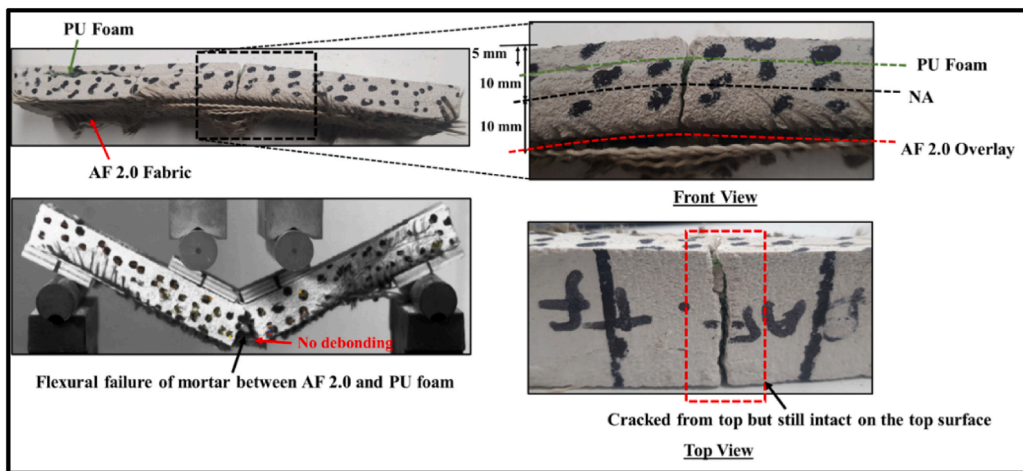


(c) Details of failure mode II of AF 2.0 overlaid composites (AF-NO) under 4-point bending test

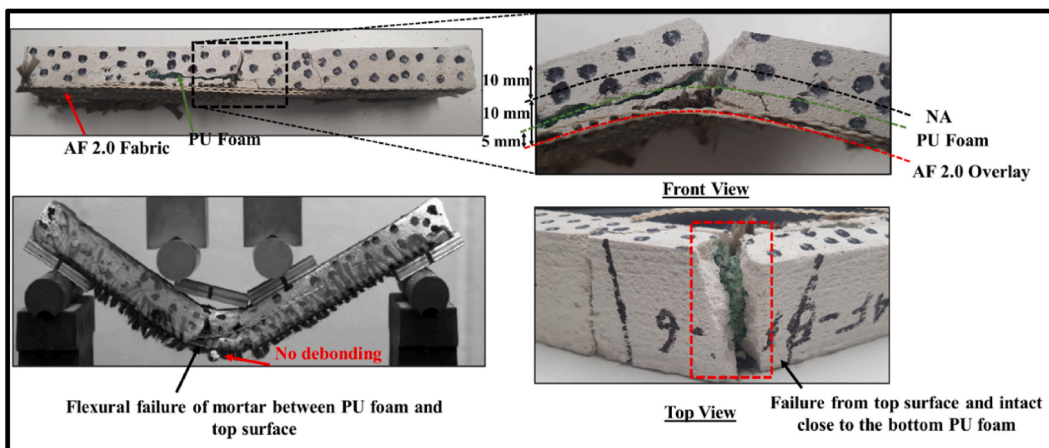
Fig. 6. Failure modes of overlaid AF 2.0 fabric composite (AF-NO) at 300 mm/min.



(a) Configuration of PU insert in the AF 2.0 overlayed composite specimen

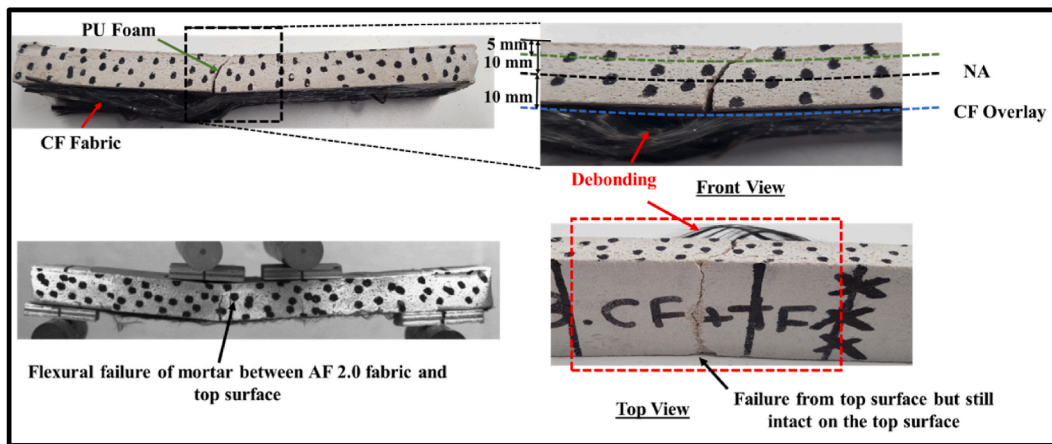


(b) AF 2.0 overlay + PU insert at the top (AF-TF)

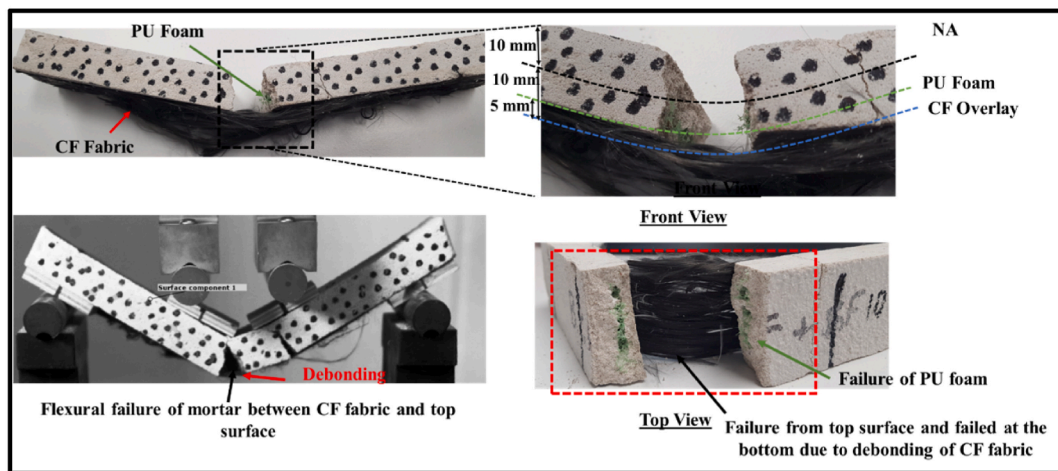


(c) AF 2.0 overlay + PU insert at the bottom (AF-BF)

Fig. 7. Failure modes of the tested PU inserted AF 2.0 overlayed specimen (AF-TF and AF-BF) at 300 mm/min.



(a) CF overlay + PU insert at the top (CF-TF)



(b) CF overlay + PU insert at the bottom (CF-BF)

Fig. 8. Failure modes of the tested PU inserted CF overlayed specimen (CF-TF and CF-BF) at 300 mm/min.

cracks widening at their locations but the PU foam fractured due to the localised stress developed when the debonding and rupture of the CF fabric occurred as shown in Fig. 9(g).

In comparison, AF 2.0 overlayed specimens showed no evidence of delamination/debonding or fracturing of the fabric (Group V, Fig. 9(h) and (i)). The PU foam inserted within the top compression face allowed the integration of the specimen without debonding of the fabric Group VI Fig. 9(j). Fig. 9(k) shows the insertion of PU foam below the neutral axis prevented the cracks widening at its location without rupturing the foam due to the NPR effect of the AF 2.0 fabric overlayed at the bottom of the specimen. The benefits of PU foam insertion were hence superior when the cementitious matrix was overlayed with the AF 2.0 fabric.

3.2. Flexural capacities of the composites

Figs. 10–14 display load-displacement curves for control specimens made of plain mortar, as well as specimens containing CF fabric with PU foam and comparable composite specimens where CF was replaced with AF 2.0 fabric. These figures illustrate the typical behaviour of these specimens. The corresponding displacement profiles of each specimen from DIC software are also provided as image inserts in these figures. The transition points (a), (b), and (c) marked on the load-displacement curves of these specimens are described here.

Point (a) – Initiation of cracking: The first crack occurred at a very low displacement corresponding to approximately 25% of peak load in all the specimens.

Point (b) – Peak load: All specimens showed a steady increase in the load capacity after initial cracking until peak load was realised. The control specimen (PM-NO) showed the lowest peak load of around 175 N; the CF overlay specimen (CF-NO) exhibited the largest load of 480 N, whilst the PU foam insertion in CF overlay composite specimens slightly reduced the peak load to 410 N. The

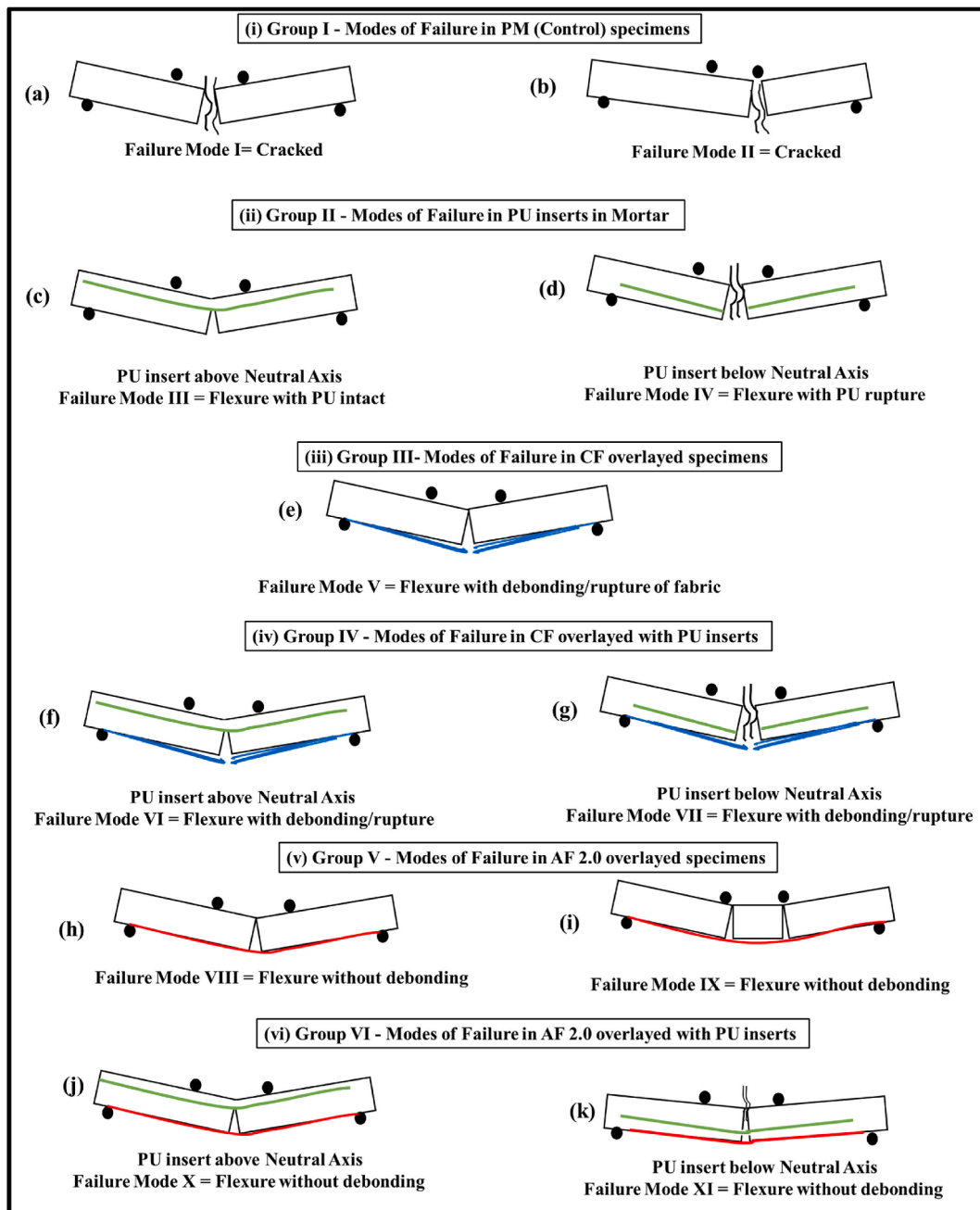


Fig. 9. Failure modes of the tested specimens.

AF 2.0 overlayed specimen (AF-NO) achieved a maximum peak load of 230 N, Whilst the PU foam insertion in AF 2.0 overlay composite specimen attained a peak load of 195 N.

Point (c)- Ultimate point: The ultimate point was defined as the points where the mortar matrix was fully damaged exhibiting clear localisations of damage with wider cracks in the tension zone and crushed mortar in the compression zone. The ultimate stages of composites are considered at 80% drop in peak load and reported in these sections. The definition of ultimate stage is consistent with the rupture of tensile layers in a brittle matrix provided by the researchers [55,56] and hence considering the ultimate stage as an 80% drop in peak load is justifiable.

The ultimate points of the CF/AF 2.0 and PU inserted in CF/AF 2.0 overlayed composite were identified when the damage was localised across the whole thickness of the mortar as marked (c) in Figs. 11–14. At the ultimate point, CF Specimens exhibited a high level of debonding and in many instances, rupture of CF fabric; none of the AF 2.0 specimens, exhibited debonding or fracture of fab-

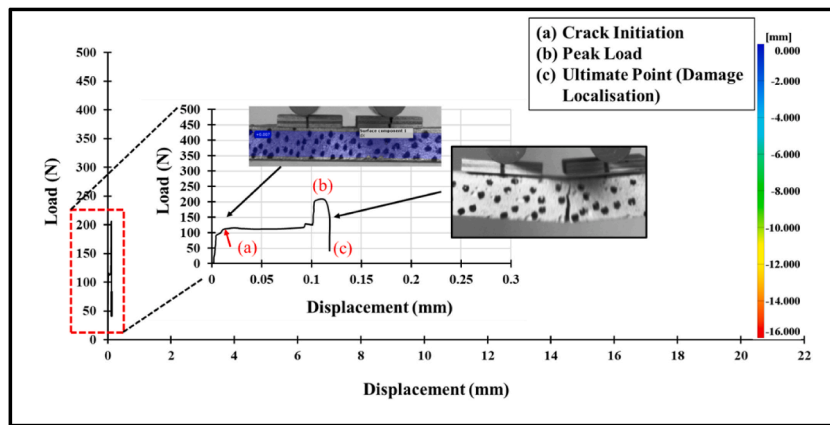


Fig. 10. Load vs Displacement curve -Control Specimen (PM-NO) at 300 mm/min.

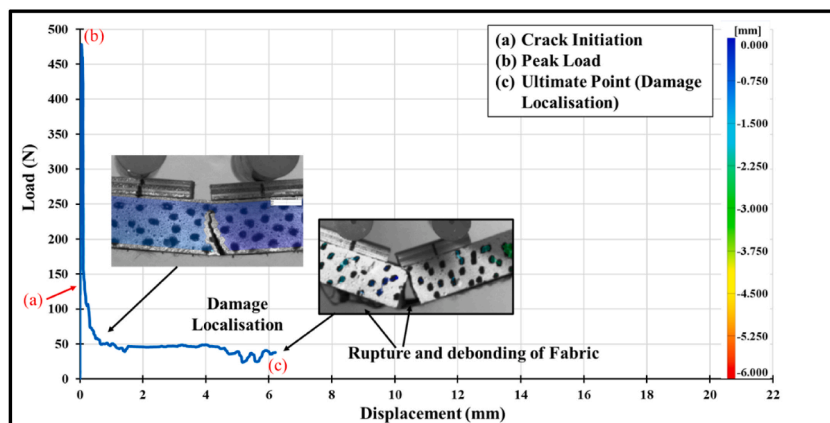


Fig. 11. Load vs Displacement curve _CF overlay specimen (CF-NO) at 300 mm/min.

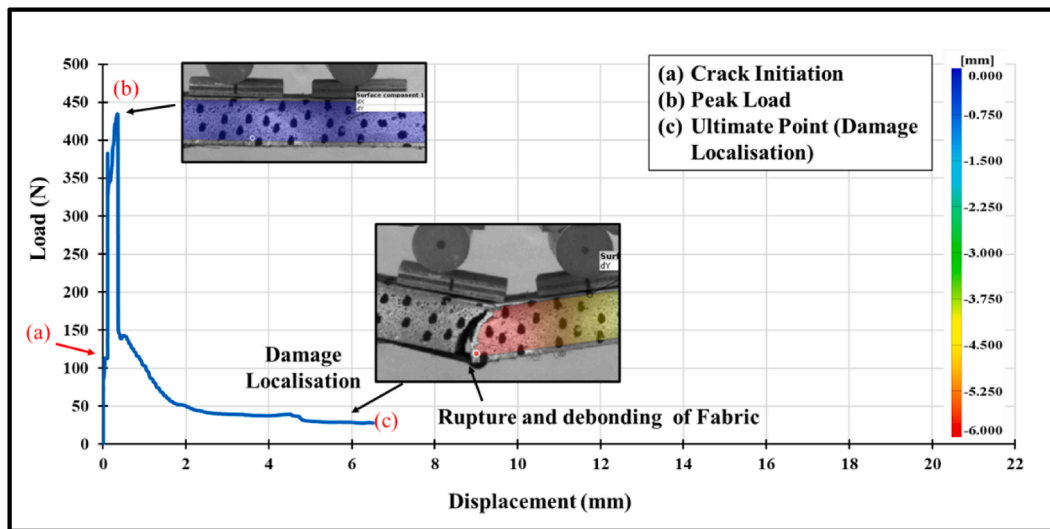
ric at the ultimate stage. As the load-displacement characteristics of the CF and AF 2.0 composite specimens shown in Figs. 11–14 respectively are significantly different, the displacements corresponding to these damage localisations states varied.

The control specimen (PM-NO: Plain mortar) failed suddenly, exhibiting a brittle load-displacement plot in Fig. 10. At a 300 mm/min rate of flexural loading, the control specimen showed a peak force of 200 N at a maximum displacement of 0.012 mm.

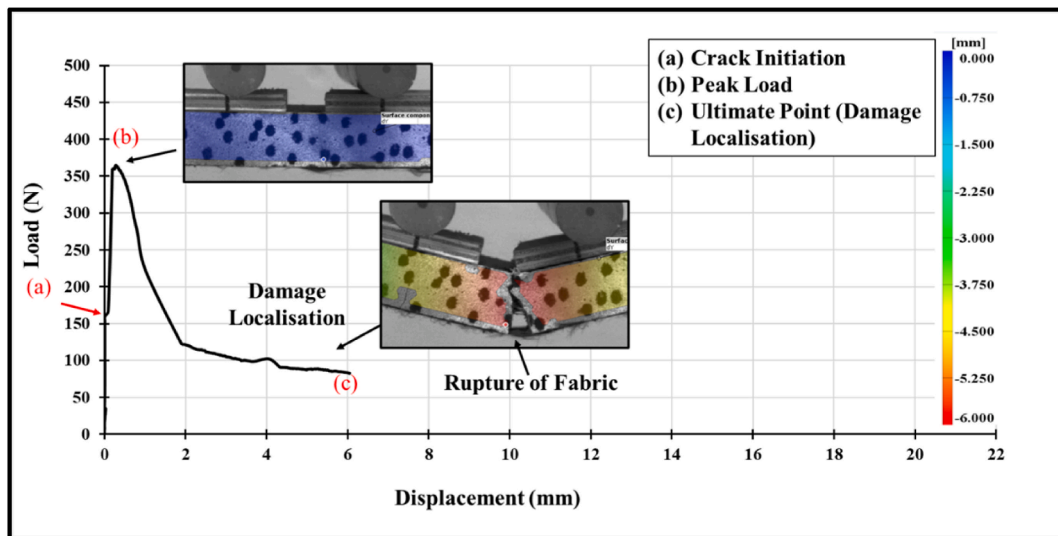
The specimens with CF overlay and PU insert also exhibited crack initiation at about 0.2 mm deflection as identified by point (a) in Figs. 11 and 12(a, b) respectively. With further progress in the widening of these cracks, both types of specimens exhibited true composite action as can be seen by the strains in the cementitious matrix, fabric, and inserted foam. The CF overlayed specimens (CF-NO) were observed to have the highest stiffness compared to all the other specimens. Hence, the CF overlayed specimens exhibited the highest peak load at around 0.6 mm deflection as shown in Fig. 11. In the post-peak regime, the CF specimens exhibited onset and growth of debonding that culminated in substantial loss of load resistance leading to the full damage localisation of the mortar matrix with no assistance from the debonded or ruptured CF fabric. At around 5 mm displacement, the CF specimen attained the ultimate stage leading to the rupture of the CF fabric. A loss of 80% of peak load provided a sufficiently accurate definition of the ultimate stage for the CF composite as identified on the load-displacement curve of the CF composite in Fig. 11.

In comparison to the specimens with CF overlay (Fig. 11), the PU insert in the CF overlay specimens showed a reduction in the peak force at around the same deflection as shown in Fig. 12 (a) and (b). The peak force reduction was substantial when the PU foam was inserted below the neutral axis. The capacity reduction was due to the replacement of 5 mm thick mortar with PU foam, whose tensile strength was only about 0.04 MPa. From the reduced values of the load-displacement curve compared to the only AF 2.0 overlayed specimen, it can be concluded that PU foam insertion above the neutral axis enabled better performance than the PU insertion below the neutral axis. Additionally, the failure modes presented in Fig. 9 (discussed in the previous section) showed that the insertion of PU foam above the neutral axis was beneficial in preventing the splitting from the top face of mortar when compared to overlayed CF composite specimens.

On the contrary, the AF 2.0 overlay specimens (AF-NO) showed a sudden drop in the capacity immediately after the peak load, reflecting the lower stiffness of the auxetic fabric; however, as debonding was not evident, these specimens regained load resistance and sustained about 67% of the peak load for a larger increase in deflection until 16 mm. This enhancement is mainly due to the intact



(a) CF overlay specimen and PU insert above the neutral axis (CF-TF)



(b) CF overlay specimen and PU insert below the neutral axis (CF-BF)

Fig. 12. Load vs Displacement curve _ PU insert CF overlayed specimens (CF-TF and CF-BF) at 300 mm/min.

bonding of the AF 2.0 fabric to the mortar, which is clearly evident in the snapshots provided in Fig. 13. The ultimate stage of the AF 2.0 overlayed composite approximately coincides with its peak stage, as shown in Fig. 15.

The AF 2.0 overlayed specimens with PU insertions either above or below the neutral axis showed similar behaviour to the CF composite specimens. They showed the same traits of reduction in the peak load capacity as seen in Fig. 14 (a) and (b), and the capacity reduction was substantial when the PU foam was inserted below the neutral axis. As before, the maximum reduction of the peak load when the PU insertion was below the neutral axis was due to the replacement of the 5 mm thick cementitious matrix by the PU foam. However, an enhanced deflection was observed when the PU foam was inserted above the neutral axis in the overlayed composite specimen. An increase in the flexural displacement of about 25% from 16 mm to 20 mm was observed. This was due to the non-debonding or non-fracturing of the AF 2.0 fabric during the experiment. Whereas, when the PU foam when inserted above the neutral axis the specimen showed almost similar flexural deflection compared to the AF 2.0 overlayed composite specimen. Considering load-displacement curves and failure modes, the PU foam inserted above the neutral axis within AF 2.0 overlayed composite specimens performed superior to PU insert below the neutral axis of the overlayed composite specimens.

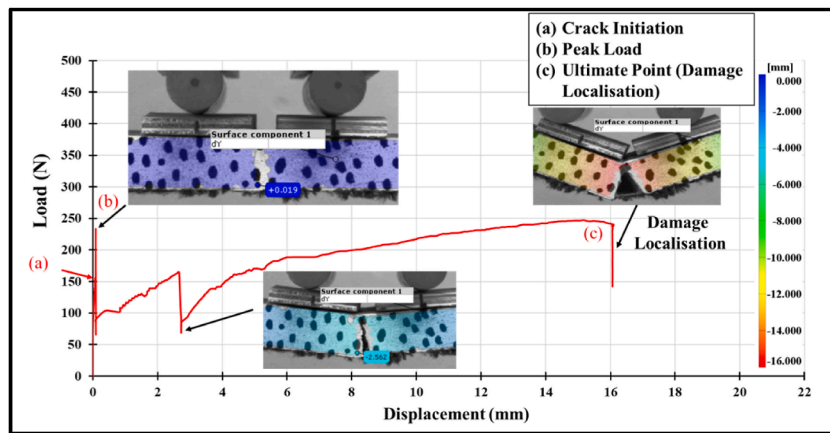


Fig. 13. Load vs Displacement curve _ AF 2.0 fabric overlay specimen (AF-NO) at 300 mm/min.

3.3. Energy absorption of the composites

Energy absorbed was determined from the area under the flexural load-displacement curve up to the ultimate point. The load-displacement characteristics of the CF and AF 2.0 composites shown in Figs. 11–14 are significantly different, and the displacements corresponding to these damage localisation states varied.

The area under the curves was calculated by multiplying the points on the curve with their corresponding X- and Y-axis units. The summation of all the multiple of the individual units provided the total energy absorption in Joules (J). For the curve shown in Fig. 11, the energy absorbed was 0.510 J. Absorbed energy for all the other specimens was determined using the same method.

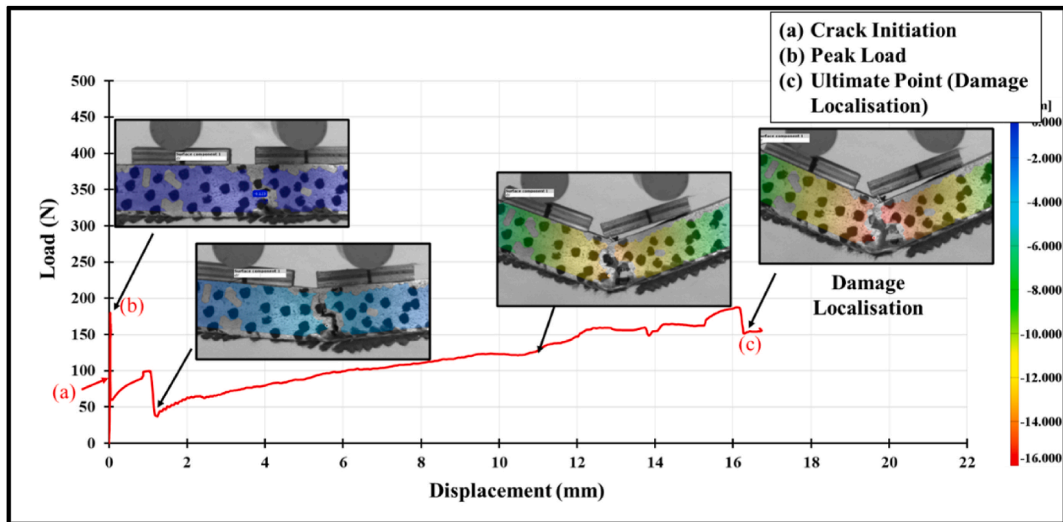
3.4. Summary of test results and discussions

Table 2 provides a list of the specimens tested and their test results. The results of the experiments are presented in columns 3–7, with mean peak load in column 3, mean load at the ultimate point in column 4, mean displacement corresponding to peak load in column 5, energy absorbed in column 6. Column 7 presents the ratio of the energy absorbed by the specimen with respect to that by the plain mortar sample. The observed mode of failure is presented in column 8. These results were discussed in detail in section 3.1 above, accompanying the failure modes discussed with reference to Fig. 9.

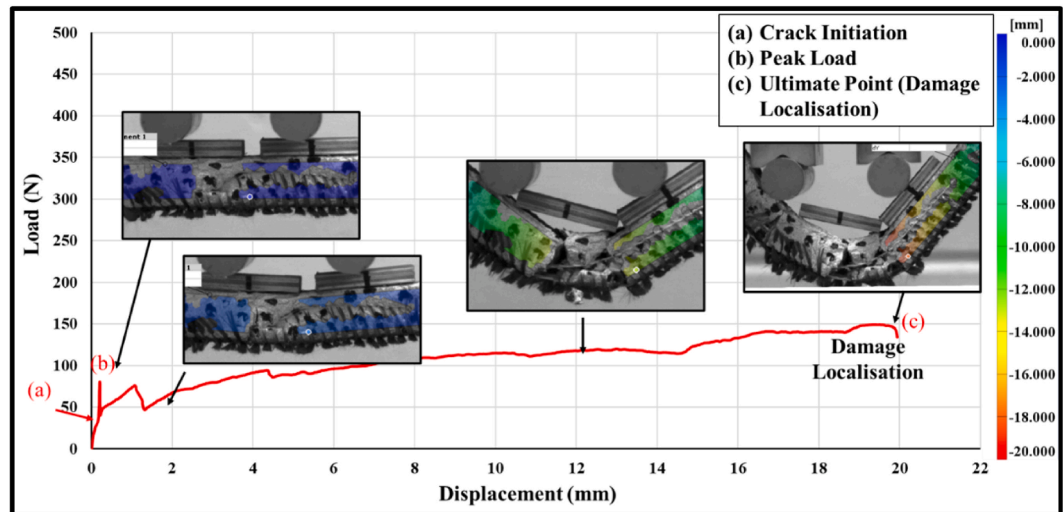
Table 2 shows that all specimens exhibited an increase in peak load with an increase in the rate of loading, consistent with the failure mechanisms of engineering materials. Specimens with AF 2.0 overlays exhibited only a marginal increase in peak load of approximately 7.5% at the lowest rate of loading (1 mm/min) to 12.5% at the highest rate of loading (300 mm/min) compared to control specimens. However, CF overlaid composite specimens showed significant enhancement of about twice the peak load compared to control specimens when tested at the low and high loading rates. These CF overlaid specimens however showed a marginal increase in peak load with a high rate of loading but failed at a lower displacement compared to AF 2.0 overlaid specimens, which affect the flexibility of the specimens significantly. The flexibility reduction within the CF composite specimens was due to bond failures and rupture of overlaid CF fabric.

The energies absorbed by all the specimens are presented in column 6 of Table 2 and show that the AF 2.0 overlaid specimens are the best-performing group. The energy absorbed was larger than the corresponding control specimens by a factor of about 147 times higher for the rate of loading of 300 mm/min. In contrast, the CF overlaid composite specimens at 300 mm/min exhibited a lower energy absorption by 45% compared to the AF 2.0 overlaid specimens. The high energy absorption of AF 2.0 was observed to be due to the non-debonding of the AF 2.0 fabric, which was caused by the negative Poisson's ratio effect of the fabric at the interface. Enhancements in the energy absorption of the specimens compared to those of the Plain mortar specimens at the different rates of loadings are presented in column 7.

The PU foam layers inserted above and below the neutral axis within the composite specimens influenced the failure modes significantly. Inserting the foam above the neutral axis was found to be better, as the foam prevents crack widening in the plain mortar when present in the compression zone. Non-splitting behaviour of the PU foam inserted at the top of the composite will be effective in preventing the spalling of mortar render from the surface of the structure during the high rate of loading. However, the load-displacement curve of the foam-inserted specimens showed a reduction in initial peak load without compromising maximum displacement or flexibility as presented in columns 3 and 5. Energy absorption calculations showed that AF 2.0 overlaid specimens with PU foam inserts above the neutral axis showed only a reduction of 6.5% at the highest rate of loading of 300 mm/min compared to the AF 2.0 overlaid specimens without the PU inserts. Similarly, about 12% reduction in energy absorption was observed in CF overlaid specimens with the PU foam inserted above the neutral axis compared to that in CF overlaid specimens without PU inserts when loaded at 300 mm/min. Enhancements in the energy absorption of the AF 2.0 overlaid composite specimens were mainly due to the higher deflections observed during the flexural bending as can be clearly visible from the curves of flexural load and energy absorption in Fig. 15(a) and (b). These two Figures briefly capture the important and interesting outcomes of the present research. Fig. 15(a) shows that the flexural loads of the CF overlaid composites are the highest compared to the other specimens. Whereas Fig. 15



(a) AF 2.0 fabric overlay specimen and PU insert above the neutral axis (AF-TF)



(b) AF 2.0 overlay specimen and PU insert below the neutral axis (AF-BF)

Fig. 14. Load vs Displacement curve _ PU insert AF 2.0 overlayed specimens (AF-TF and AF-BF) at 300 mm/min.

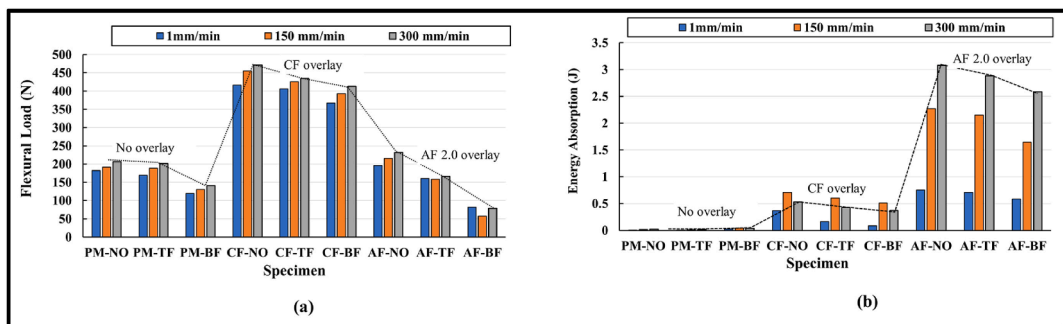


Fig. 15. (a) Flexural Capacity and (b) Energy absorption of the composites at different loading rates.

Table 2
Summary of results for specimen groups.

Specimen group	Rate of loading (mm/min)	Mean peak load in (N) (COV %)	Mean load (N) at ultimate point	Mean deflection at peak load (mm)	Absorbed energy (Avg.) J,	Ratio Energy absorption (Specimen/Ref)	Failure Shape (Refer Fig. 9)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
PM-NO-001-D	1	182.36 (0.7)	0.00	0.10	0.0047	Ref	(a)
PM-TF-001-D		168.95 (0.49)	32.42	0.10	0.0037	0.78	(c)
PM-BF-001-D		119.23 (0.23)	13.73	0.36	0.023	4.89	(d)
CF-NO-001-D		415.90 (1.31)	42.30	1.77	0.370	78.72	(e)
CF-TF-001-D		405.67 (0.20)	38.67	1.86	0.165	35.10	(f)
CF-BF-001-D		367.13 (1.56)	24.89	2.10	0.085	18.08	(g)
AF-NO-001-D		195.88 (1.20)	169.75	12.16	0.752	160.01	(h)
AF-TF-001-D		160.61 (2.41)	165.12	11.12	0.706	150.21	(j)
AF-BF-001-D		81.96 (3.29)	127.19	17.51	0.586	124.68	(k)
PM-NO-150-D	150	191.86 (1.92)	0.00	0.11	0.016	Ref	(a), (b)
PM-TF-150-D		188.43 (1.31)	8.90	0.13	0.012	0.75	(c)
PM-BF-150-D		130.41 (0.30)	26.31	0.61	0.047	2.94	(d)
CF-NO-150-D		455.11 (0.52)	56.14	3.71	0.710	44.37	(e)
CF-TF-150-D		425.02 (1.09)	52.14	3.91	0.606	37.87	(f)
CF-BF-150-D		392.63 (1.65)	22.65	4.12	0.513	32.06	(g)
AF-NO-150-D		215.64 (0.96)	193.38	15.28	2.270	141.87	(h), (i)
AF-TF-150-D		158.26 (3.12)	188.78	15.27	2.151	134.44	(j)
AF-BF-150-D		57.23 (3.46)	137.36	18.42	1.645	102.81	(k)
PM-NO-300-D	300	206.21 (0.06)	0.00	0.12	0.021	Ref	(a), (b)
PM-TF-300-D		201.41 (2.22)	2.77	0.12	0.018	0.86	(c)
PM-BF-300-D		140.47 (1.05)	45.26	0.43	0.039	1.86	(d)
CF-NO-300-D		471.25 (0.56)	63.12	6.12	0.530	29.44	(e)
CF-TF-300-D		434.48 (0.59)	48.14	6.21	0.425	20.23	(f)
CF-BF-300-D		412.43 (0.80)	31.24	6.55	0.370	17.61	(g)
AF-NO-300-D		231.91 (2.25)	237.48	16.08	3.081	146.71	(h), (i)
AF-TF-300-D		165.67 (2.49)	192.23	15.72	2.882	137.23	(j)
AF-BF-300-D		78.89 (4.89)	138.23	20.37	2.584	123.04	(k)

(b) shows that the energy absorptions (calculated as the area under the flexural load-displacement curves) are highest for the AF overlaid specimens. The reason is obvious and as already discussed above is due to the non-debonding characteristic of the AF 2.0 fabric, which significantly enhances the flexural deflections of the specimens.

The results show that the Auxetic cementitious composites were significantly efficient in energy absorption compared to the CFRP cementitious composites. Results were compared with the behaviour of textile-reinforced cementitious composites from the recent literature to evaluate the effectiveness of the auxetic cementitious composites. The comparisons were made in terms of energy absorption enhancements of the composite specimens with respect to the control specimens. For example, in Fig. 15 (b), the energy absorbed by the specimen AF-NO at 300 mm/min loading rate is 3.081 J, and for the control specimens PM-NO at the same rate was observed to be only 0.021 J. The auxetic overlay increased the energy absorption by about 146.7 times higher than that of the control PM-NO specimens and about 6 times higher than the CF overlaid specimens when tested at a higher loading rate. Similar ratios of energy ab-

sorbed by textile-reinforced cementitious composites reported in the literature were calculated and are summarised in Fig. 16. The energy enhancement factors of all tested textile-reinforced cementitious composites under flexural bending are plotted on the x-axis of the graph presented in Fig. 16. Du et al., 2021 [57] proposed a new composite of polyvinyl alcohol fibre-reinforced ferrocement cementitious composites, which were tested under 4-point flexural loading. This composite was a cement matrix reinforced with PVA fibres and steel wire mesh of different grid sizes. It can be seen from Fig. 16 that the energy efficiencies are in the range of 5–7 for these composites as compared to plain mortar. Xu et al., 2021 [58] customised the lattice structures with different unit cell size combinations using 3D printing and calculated the maximum energy absorption capabilities of the 3D-printed specimens which was only about 13 times compared to the control specimen (shown in Fig. 16). In 2018, Quadflieg and colleagues attempted to enhance the energy absorption capability of warp-knitted fabrics [59]. They achieved this by coating the fabrics with various polymers, including potassium silicate, epoxides, and carboxylated styrene butadiene rubber, prior to insertion into the cement matrix. These coatings resulted in a significant increase in energy absorption capacity, from 9 times to 11 times, and eventually by 27 times as shown in Fig. 16. Similarly, Pakravan et al. [60] also improved the performance of cementitious composites by reinforcing with polyvinyl alcohol-polypropylene hybrid fibres and achieved significant enhancement of about 108 times higher than the control specimens. The enhancement was due to adding 2% of polypropylene fibers, which significantly enhanced the flexural strength. However, it isn't easy to achieve when embedded as a textile or a fabric of same fibers as a reinforcement due to the debonding/delamination issue which is also reported for the CF fabric in this current paper. Halvaei et al. [61] in 2014 enhanced the flexural and energy absorption capability of the cementitious matrix by reinforcing with three different high modulus fibres such as Acrylic, polypropylene and nylon 66 and the maximum enhancement in energy absorption due to nylon 66 insertion was about 66 times higher than the cementitious matrix. Similarly, other researchers, including De Munck et al. [62], Peled et al. [63], Padalu et al. [64], Khattak et al. [65], and Martins et al. [66], also attempted to enhance the flexural and energy absorption capabilities of cementitious composites by reinforcing them with high modulus fibres. These fibres are typically composed of materials such as AR glass, polyethylene, basalt, polyparaphenylene ben-

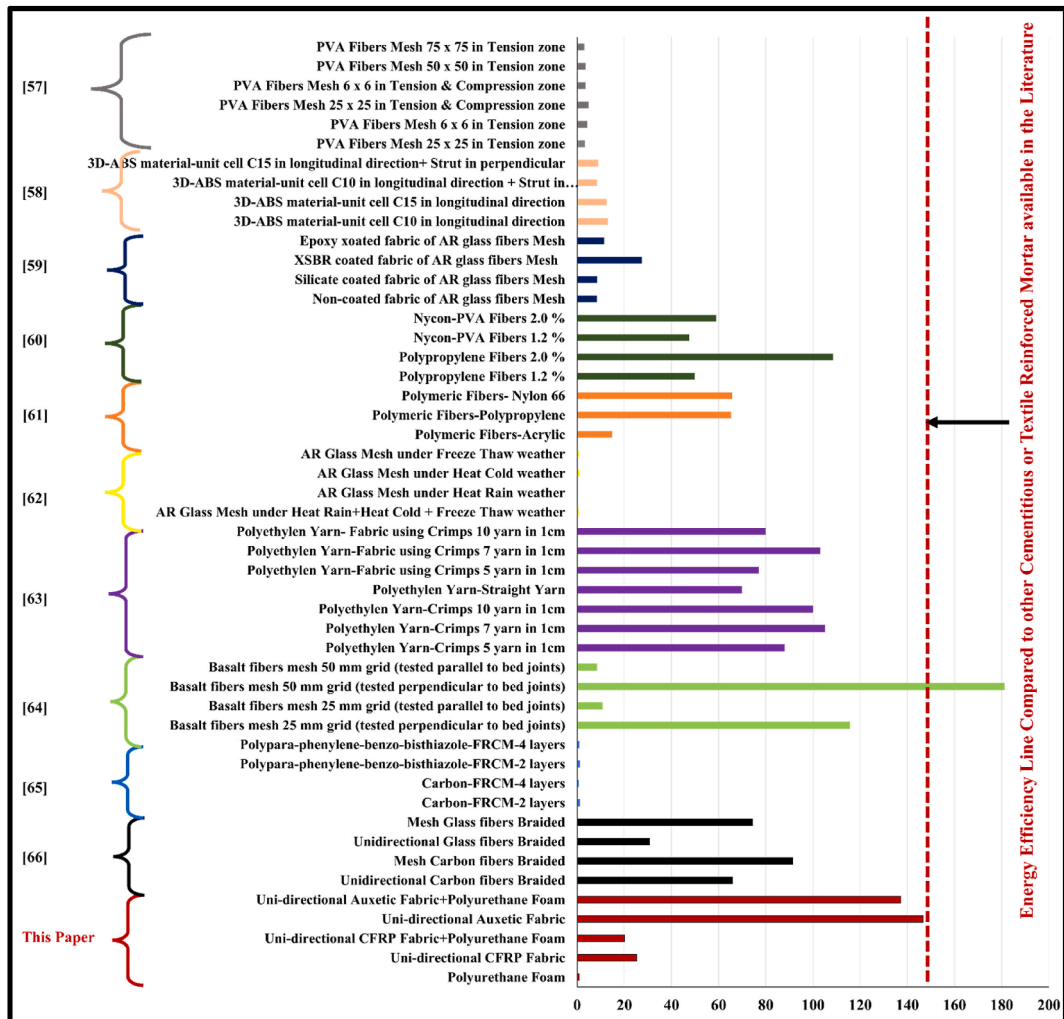


Fig. 16. Comparison of energy enhancement factors of different specimens under flexure with its control or referenced specimens (Reported from Literature [57–66]).

zobisoxazole, glass, and carbon. However, upon comparing the energy enhancement ratios of these specimens, it was observed that only basalt fibres/mesh or carbon fibres/mesh were able to absorb energy up to approximately 181 times or 91 times, respectively, as illustrated in Fig. 16. This exceptional energy absorption was attributed to the very high flexural strength of the basalt or carbon fibres, which was comparable to the 146 times reported for the auxetic fabric insertion in cementitious composites discussed in this paper. Nevertheless, the use of basalt fibres and CFRP fibres in cementitious composites was found to result in lower ductility and was considered non-economical and non-environmentally friendly. The histogram filled with red colour in Fig. 16 represents the enhancements in the energy absorption for the specimens tested in the current paper. The red dotted vertical line clearly shows that the energy absorption capability under flexural bending for the auxetic cementitious composites is significantly higher compared to other specimens.

4. Conclusions

A comprehensive experimental investigation on the flexural response of cementitious composites reinforced with polyurethane foam and auxetic fabric layers was carried out in this paper. The main aim of the study was to compare the energy absorption characteristics of the developed composites for their application as renders in impact mitigation of structures. Experimental testing was carried out at three rates of loading, viz., 1, 150 and 300 mm/min using four-point bending tests. Failure mode, peak load, load-displacement behaviour, and energy absorption were evaluated from the test data of 90 specimens. Results showed that the use of overlaid Carbon fibre (CF) and auxetic (AF 2.0) fabric, along with the insertion of PU foam above the neutral axis, improved the energy absorption characteristics of the cementitious matrix renders when tested at various loading rates. Comparisons of energy absorption capabilities with textile-reinforced cementitious composites (treated in the recent literature) were also made. The following conclusions have emerged from the study.

- Auxetic fabric 2.0 overlaid specimens (AF-NO) showed a significant reduction in debonding and enhanced energy absorption characteristics of the cementitious matrix. The superior performance of AF 2.0 cementitious composites is attributed to its negative Poisson's ratio.
- AF 2.0 overlaid (AF-NO) and CF overlaid composites (CF-NO) consistently exhibited enhanced response of peak load, displacement, and energy absorption with the increase in the rate of loading. However, the energy dissipated by the AF 2.0 specimens was about 147 times higher and for the CF specimens, it was about 25 times higher than that of controlled specimens at 300 mm/min loading rate.
- The PU foam insertion above the neutral axis in the overlaid composite specimens enhanced the composite action of the specimens by eliminating the crack widening through the thickness of mortar above the neutral axis. In particular, the AF 2.0 overlaid cementitious composite specimens without the PU inserts displayed crack widening up to the top surface to allow splitting through the thickness of the specimen.
- CF overlays cementitious composite specimens (CF-NO) exhibited debonding and fibre fracturing. Although it significantly contributed to the peak load, its energy absorption was rather low (less by 45%) compared to AF 2.0 overlays cementitious composite specimens.
- Compared to other cementitious composites reported in the literature with various strengthening fibres and fabrics, the auxetic cementitious composites tested in this study with and without PU foams demonstrated an enhancement in the energy absorption under bending.

This research is significant in terms of advancing the knowledge of protective composites for wall structures to save them from vehicular crashes. The findings of the present study reveal that AF 2.0 overlaid composite specimens, with PU foam insertions in the compression zone above the neutral axis, and Auxetic 2.0 fabric overlaid in the tension zone below the neutral axis, exhibit promising potential as effective impact resistance solutions for buildings and wall structures vulnerable to vehicular impacts. To achieve this impact resistance, the PU foam needs to be rendered using a mortar matrix on the front face of the wall, which corresponds to the compression zone above the neutral axis of the composite wall structure. On the other hand, the Auxetic 2.0 fabric can be applied to the back face of the wall, corresponding to the tension zone below the neutral axis of the composite wall structure. The incorporation of PU foam not only minimizes the splitting of the mortar render in the compression zone of the composite wall structure but also complements the strength provided by the auxetic fabric on the bottom tension face of the composite wall structure. This combined approach effectively enhances the impact resistance of the wall structure. In conclusion, the strategic use of foam insertions and auxetic fabric overlays can help prevent the penetration of an impactor at the front face of a masonry wall structure, making it a valuable solution for enhancing the structural integrity of buildings and walls in the face of potential vehicular impacts. Future studies will present more conclusive evidence on the composite wall structure's performance.

Credit authorship contribution statement

Mohammad Asad: Conceptualization, Writing – original draft, Data curation, analysing, Nay Win: Data analysing. Tatheer Zahra: Writing – review & editing. David Thambiratnam: Chief Investigator of the Project, Funding acquisition, Writing – review & editing, Tommy Chan: Chief Investigator of the Project, Funding acquisition, Review & editing and Yan Zhuge: Chief Investigator of the Project, Funding acquisition, Review & editing.

Declaration of competing interest

The authors of this paper certify that they have no affiliations with any organisation with any financial or non-financial interest in the subject matter or materials discussed in this manuscript.

Data availability

Data will be made available on request.

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